2017-DSE
BIO
PAPER 1B

B

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Candidate Number									

HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY

HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2017

BIOLOGY PAPER 1

SECTION B : Question-Answer Book B

This paper must be answered in English

INSTRUCTIONS FOR SECTION B

(1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1, 3, 5, 7 and 9.

(2) Refer to the general instructions on the cover of the Question Paper for Section A.

(3) Answer **ALL** questions.

(4) Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.

(5) Supplementary answer sheets will be supplied on request. Write your candidate number, mark the question number box and stick a barcode label on each sheet, and fasten them with string **INSIDE** this Question-Answer Book.

(6) Present your answers in paragraphs wherever appropriate.

(7) The diagrams in this section are **NOT** necessarily drawn to scale.

(8) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

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SECTION B

Answer **ALL** questions. Write your answers in the spaces provided.

1. For each of the components of the musculoskeletal system listed in Column I, select from Column II one phrase that correctly describes it. Put the letter in the space provided. (3 marks)

Column I		*Column II*
Ligament	______	A. Inelastic tissue found at the two ends of a skeletal muscle
Tendon	______	B. Elastic tissue found at the two ends of a long bone
Cartilage	______	C. Inelastic tissue that surrounds a joint
		D. Elastic tissue that binds bones together

2. Denise ate a piece of pineapple preserved in a sugar solution and noticed that it was softer than fresh pineapple. Explain this phenomenon. (3 marks)

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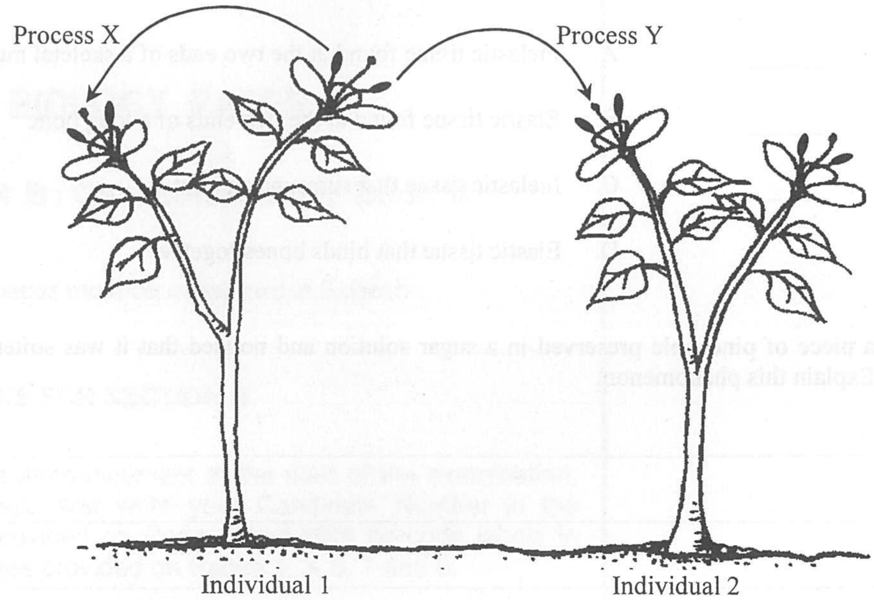
3. The diagram below shows certain processes occurring in a species of flowering plant:



(a) What is process Y? (1 mark)

(b) Describe the sequence of events leading to fertilisation after process Y has completed. (4 marks)

(c) Explain briefly why process Y is better than process X in terms of evolution. (2 marks)

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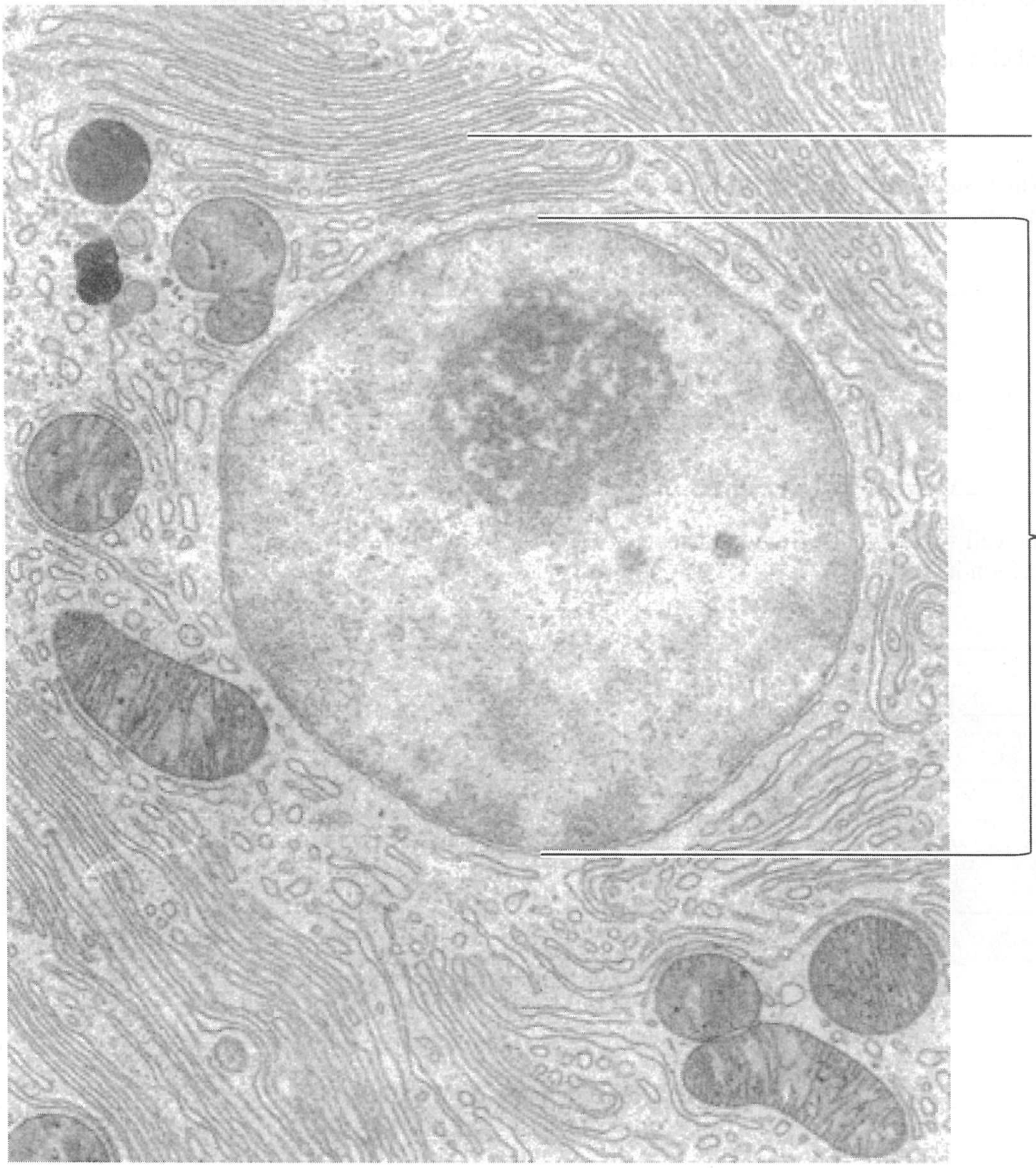

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4. The electron micrograph on the opposite page shows some structures of a human cell.

(a) Label A and B. (2 marks)

A: ______________________ B: ______________________

(b) Which stage of the cell cycle is shown in this photomicrograph? Give a reason to support your answer. (2 marks)

(c) The cell was obtained from the pancreas. How do A and B work together such that this cell can perform its function? (4 marks)

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5. (a) Briefly describe the breathing actions that bring air into the lungs. (4 marks)

(b) In the following diagrams, Diagram I shows some structures of a human lung while Diagram II shows a collapsed lung if it is ruptured at location Y:

Diagram I

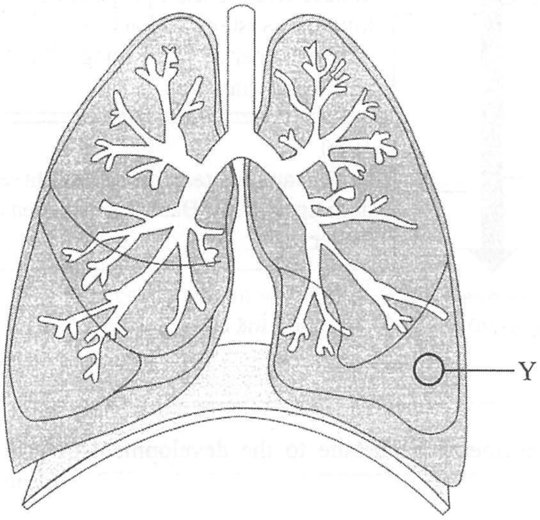
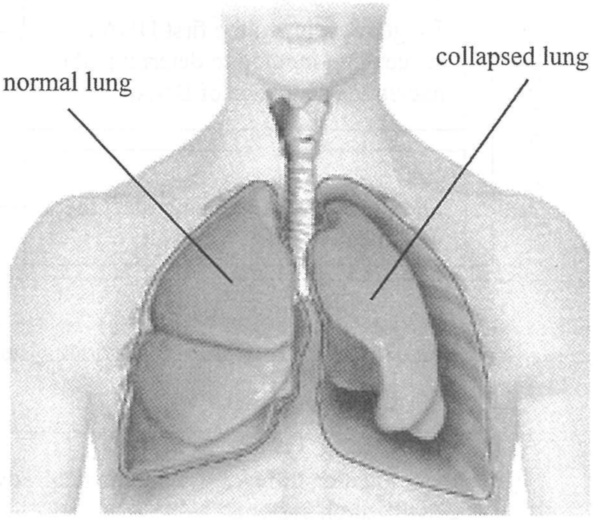


Diagram II



Explain why the lung collapses if it is ruptured at location Y. (2 marks)

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6. The chart below shows the timeline of some major advancements of technology and the development of different classification systems:

	Advancements of technology	*Development of different classification systems*
Microscopy	1660 Hooke observed cork cells using a simple microscope.	
		1735 Linnaeus proposed the two kingdom system: Vegetabilia and Animalia
Microscopy	1886 The compound light microscope was invented.	
		1937 Chatton proposed the two empire system: Prokaryota and Eukaryota
DNA sequencing	1975 Sanger developed the first DNA sequencing method to determine the nucleotide sequence of DNA.	
		1977 Woese and his team proposed six kingdom system: Eubacteria, Archaebacteria, __X__, Fungi, Plantae and Animalia
DNA sequencing	1987 The first DNA sequencing machine was invented.	
		1990 Woese and his team proposed three domain system: Bacteria, Archaea and Eukarya

(a) Name kingdom X in the six kingdom system proposed by Woese and his team in 1977. (1 mark)

(b) How did the following technological advancements contribute to the development of different classification systems? (4 marks)

Microscopy:

DNA sequencing:

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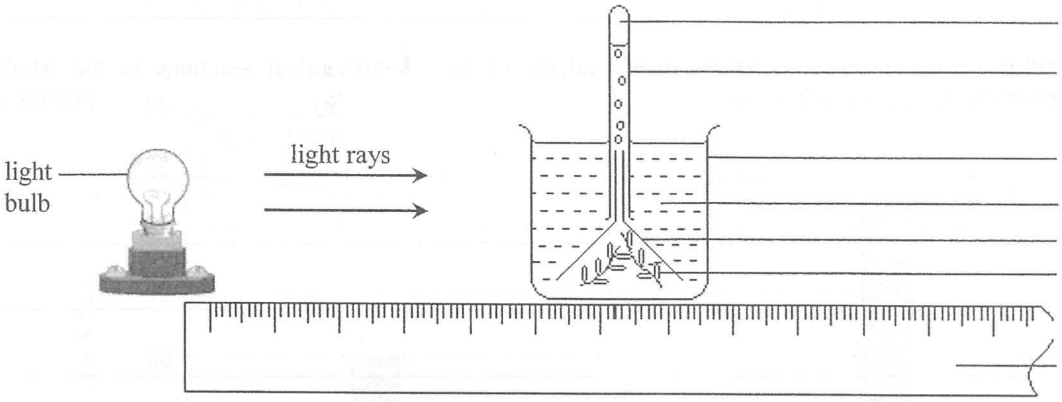
7. The diagram below shows an experimental set-up for investigating the effect of light intensity on the rate of photosynthesis:



(a) What is the assumption behind using the volume of oxygen released per unit time to indicate the photosynthetic rate? Explain your answer. (2 marks)

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(b) Suggest ***one*** modification to this experimental set-up to make sure that the result is due to the independent variable only. Explain your answer. (3 marks)

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(c) What is the significance of the ***two*** products of the photochemical reactions to the whole photosynthetic process? (4 marks)

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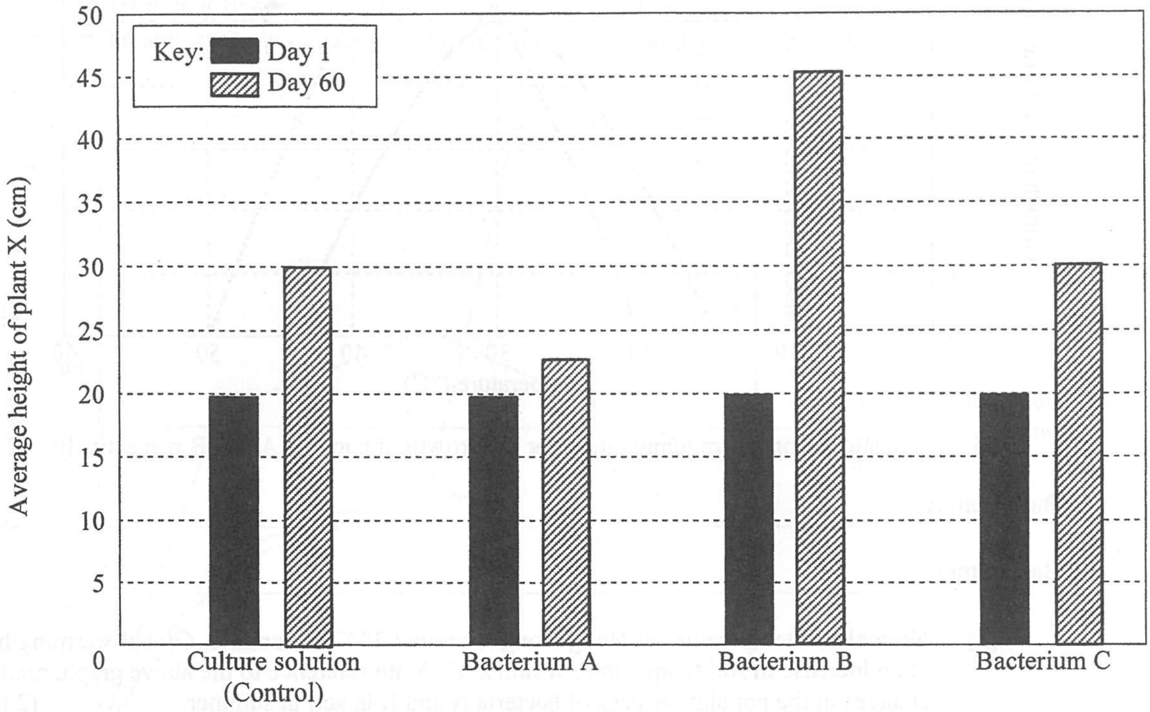
8. In an investigation about the effect of soil bacteria on the growth of plant X, three types of soil bacteria (A, B and C) were grown separately in one type of culture solution. After this, each bacterial culture was added to the soil of separate pots of plant X. A control was prepared by adding the culture solution only. The average height of plant X was recorded at day 1 and day 60 of the experiment. The results are shown in the chart below:



(a) With reference to the above results, state the effect of each type of bacterium on the growth of plant X. (3 marks)

Bacterium A: ______________________________

Bacterium B: ______________________________

Bacterium C: ______________________________

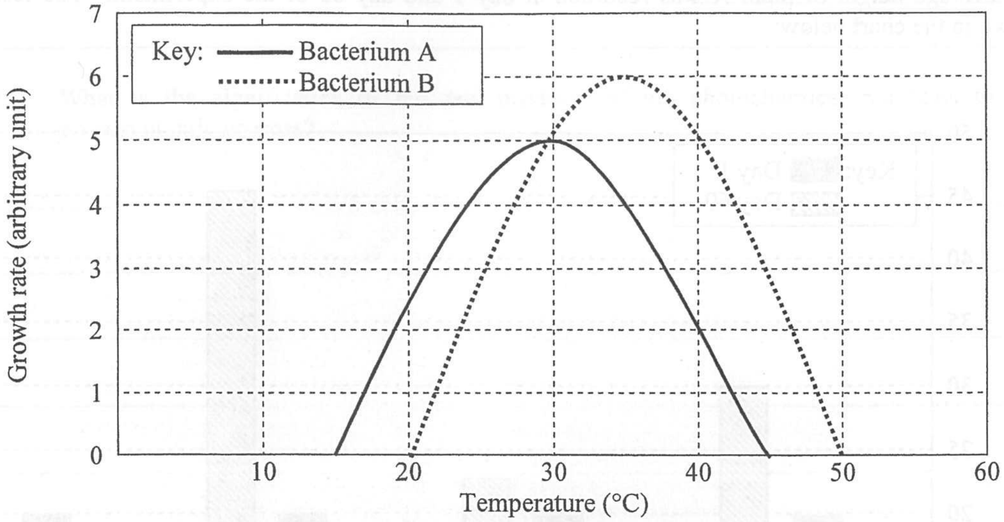
(b) It is known that both bacteria A and B are able to colonise the root of plant X and obtain nutrients from the root for growth. Suggest the possible ecological relationships between each type of bacterium and plant X. (2 marks)

Bacterium A: ______________________________

Bacterium B: ______________________________

Answers written in the margins will not be marked.

(c) In a subsequent experiment, the effect of temperature on the growth of bacteria A and B was tested and the results are shown in the graph below:



(i) Indicate the optimum temperature for the growth of bacteria A and B respectively. (1 mark)

Bacterium A: ....

Bacterium B: ....

(ii) Normal soil temperature in Hong Kong is around 30°C in summer. Global warming has led to an increase in soil temperature within 2°C. With reference to the above graph, predict the changes in the population sizes of bacteria A and B in soil in summer. (2 marks)

(d) Plant X is a foreign species that is more competitive than the native plant species in Hong Kong. With reference to your answers in (a) and (c)(ii), suggest a possible impact of global warming on the native plant community. Explain your answer. (2 marks)

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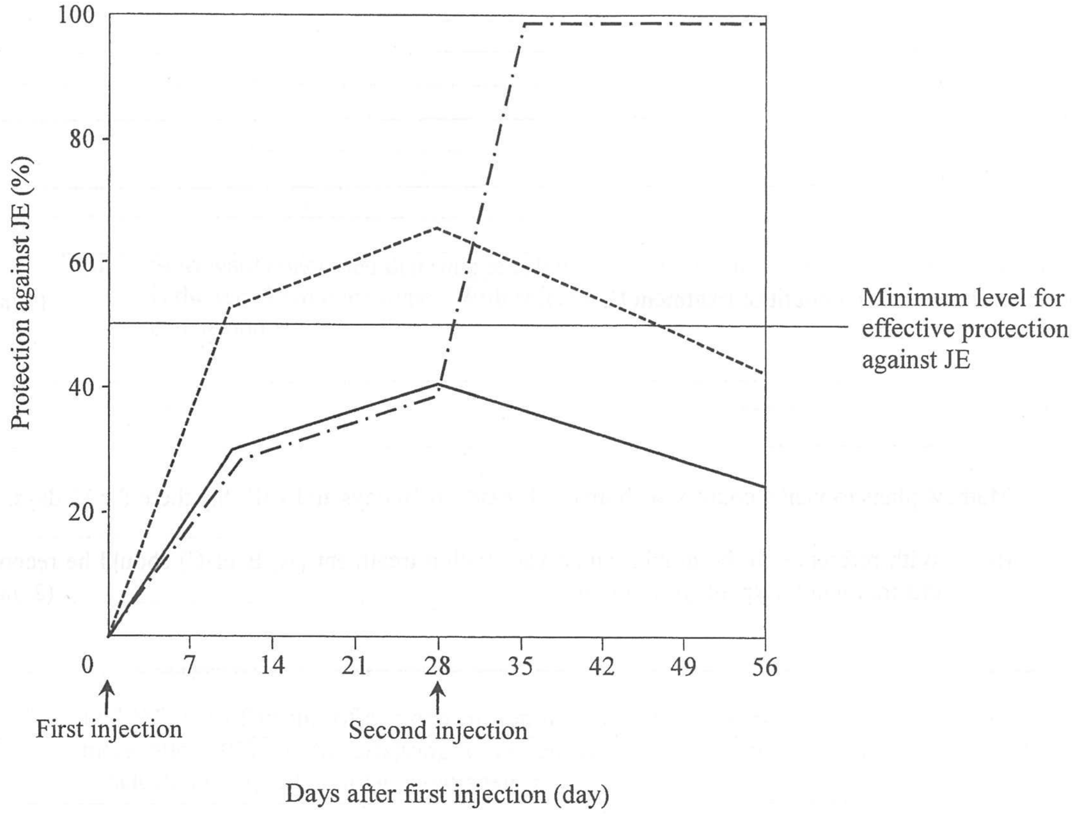
9. Japanese encephalitis (JE) is an inflammation of the brain caused by a viral infection. Scientists have developed a vaccine against the JE virus. In a study of the effectiveness of the vaccine, three groups of healthy people received different vaccination treatments and the level of protection against JE was monitored over a period of time. The results are shown in the graph below:

Key:
Treatment A: one injection of 6 μg vaccine on day 0
Treatment B: one injection of 12 μg vaccine on day 0
Treatment C: two injections of 6 μg vaccine each on day 0 and day 28 respectively



(a) What is the vector for transmitting the JE virus? (1 mark)

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(b) For treatment C, explain why there is a sharp rise in protection against JE from day 28 to day 35. (4 marks)

(c) Give ***one*** more benefit of treatment C. (1 mark)

(d) Mathew plans to visit a country with many JE cases in 10 days and will stay there for 15 days.

(i) With reference to the graph, which vaccination treatment (A, B or C) should he receive at this moment? Explain your answer. (2 marks)

(ii) As a responsible citizen, Mathew will continue to use insect repellant as a precaution for two weeks after he is back from that country. Suggest a rationale for this precaution. (1 mark)

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10. (a) In 1940, scientist Alfred Sturtevant hypothesised that the ability to roll one's tongue is determined by a single gene. His hypothesis was based on the data below:

Case	Characters of parents	Tongue rolling offspring	Non-tongue rolling offspring
I	tongue rolling x tongue rolling	28	5
II	tongue rolling x non-tongue rolling	33	22

(i) Does the trait of tongue rolling ability show continuous or discontinuous variation? Explain your answer. (2 marks)

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(ii) Sturtevant concluded that tongue rolling is the dominant phenotype while non-tongue rolling is the recessive phenotype. With reference to the above table, explain how he arrived at this conclusion. (2 marks)

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(b) In 1965, the offspring of a group of non-tongue rolling parents were studied. It was found that more than 30% of the offspring were tongue rollers. Does this finding support Sturtevant's conclusion in (a)(ii)? Explain your answer. (2 marks)

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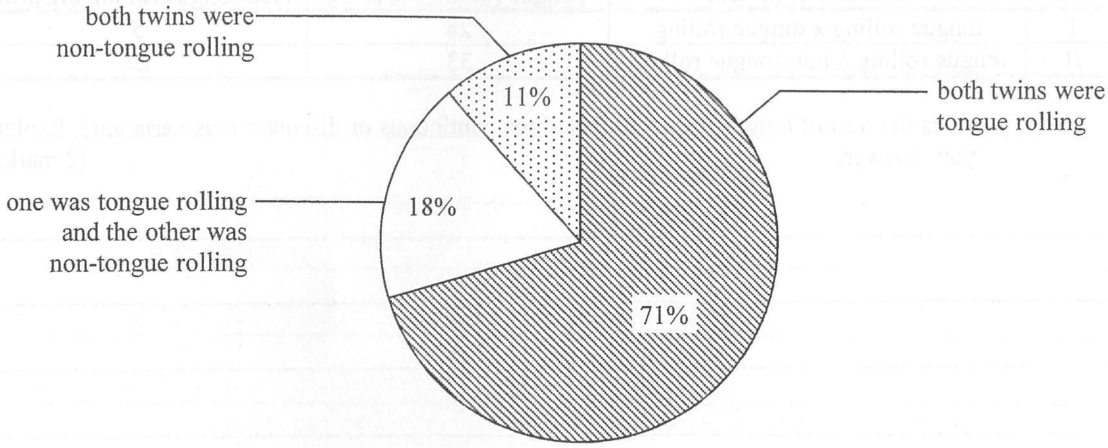
(c) In 1971, another study on identical twins was carried out to further explore the factors influencing the tongue rolling trait. The results are summarised in the chart below:



(i) What is the advantage of using identical twins as the subjects for the study? (2 marks)

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(ii) With reference to the above chart, complete the following table with data that support the conclusions. (2 marks)

Conclusion	Evidence
Genetic factors play a significant role in the determination of the tongue rolling trait.	
There are other factors influencing the tongue rolling trait.	

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(d) (i) In the above case regarding the development of knowledge about the inheritance of the tongue rolling trait, which of the following ideas about science is demonstrated? (2 marks)

Ideas about science	Put a '✓' in the appropriate spaces below
Science is a process of ongoing inquiries.	
Science is affected by social and cultural factors.	
Scientists may not arrive at the same conclusions about the same set of data.	
Scientific investigations may not require doing experiments in laboratories.	

(ii) Elaborate on how the development of knowledge about the inheritance of the tongue rolling trait can be used to demonstrate that scientists have to be open-minded. (1 mark)

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For the following question, candidates are required to present their answer in essay form. Criteria for marking will include relevant content, logical presentation and clarity of expression.

11. Some natural therapists claim that applying pressure along one's limbs toward the body trunk can improve the circulation of lymph and result in reduced body weight. However, the effects of this treatment are controversial. Briefly describe how lymph is formed from blood and returned to the blood circulatory system. For each of the claims above, discuss whether it is scientifically valid. (11 marks)

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END OF PAPER

Sources of materials used in this paper will be acknowledged in the booklet *HKDSE Question Papers* published by the Hong Kong Examinations and Assessment Authority at a later stage.

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